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Department of Civil and Mechanical Engineering
ME486/5586- Applied Finite Element Analysis
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Title: Final Group Project

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Monday, December 8th, 2025

Abstract

Title: Additive Manufactured Support Stand for Tiny House Applications: Finite Element Design and Validation

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This report presents the conceptualization, design, and finite element analysis of a lightweight support stand for Tiny House structures, developed using additive manufacturing principles. The design process began with 5 CAD concepts modeled using SolidWorks focused on minimizing weight while maintaining structural integrity under axial, lateral, and torsional loads. After evaluating each concept, the Tripod Truss concept design was selected for simulation due to its optimal balance of stiffness, mass efficiency, and manufacturability. Furthermore, this concept design incorporates fixed mounting interfaces at the top and base, as specified by the client, and was analyzed using simulation tools such as **Abaqus CAE** and **Fusion360** to validate performance under the prescribed loading conditions.

Simulation results confirmed that the selected design meets all functional requirements: it supports an axial load of **2000 N**, resists lateral forces of **500 N**, and withstands a twisting moment of **100 N·mm**. The first natural frequency was found to be ~**200.32 - 211.941 Hz**, almost exceeding the minimum resonance threshold of **200 Hz**, and the total weight of the structure was calculated to be **53.91 N**, remaining way below the **120 N** limit. These results demonstrate the viability of the proposed structure for real-world deployment and highlight the advantages of lattice-reinforced conical geometries in additive manufacturing.

Keywords: Finite Element Analysis, Additive Manufacturing, Abaqus CAE, Lightweight Structures, Tiny House Support, Resonance Frequency, Lattice Design, Aluminum 6061-T6, Structural Optimization, CAD Modeling, Modal Analysis

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Overview

The client has commissioned the development of a structurally sound and lightweight support stand for Tiny House applications, to be manufactured using additive manufacturing techniques. The design must be integrated into two fixed components, which are a mounting ring and a ground base pad, while meeting a series of functional requirements. These include withstanding an axial dead load of **2000 N**, a lateral load of **500 N**, and a twisting moment of **100 N·mm**. Additionally, the final structure must withstand a first natural frequency above **200 Hz** and maintain a total weight below **120 N**. The mounting interfaces are provided and must remain unaltered, serving as boundary conditions for all design iterations.

In response to these specifications, the engineering team has developed up to 5 conceptual CAD models that prioritize weight minimization while preserving mechanical integrity. Each concept was evaluated based on its geometric efficiency, manufacturability, and potential to meet the performance criteria. After comparative analysis, one design was selected for detailed finite element analysis. This configuration was chosen for its favorable stiffness-to-weight ratio and compatibility with additive manufacturing constraints.

Subsequent simulations were conducted to validate the selected design against the prescribed loading and resonance conditions. Modal analysis, static structural evaluation, and mass estimation were performed using industry-standard FEA software. The results of these simulations are presented in the following sections, along with a discussion of design rationale, performance metrics, and compliance with client requirements. The report concludes with recommendations for future refinement and potential deployment of the support stand in Tiny House installations.

Scope of Work

Geometry and parameters

Material: Aluminum AlSi10Mg

Mechanical Properties:

- Density:
  - $2700 \frac{kg}{m^3} \rightarrow 0.00000267 \frac{kg}{mm^3}$
- Yield Strength:
  - $70\ GPa \rightarrow 70{,}000 \frac{N}{mm2}$
- Poisson's Ratio:
  - 0.33

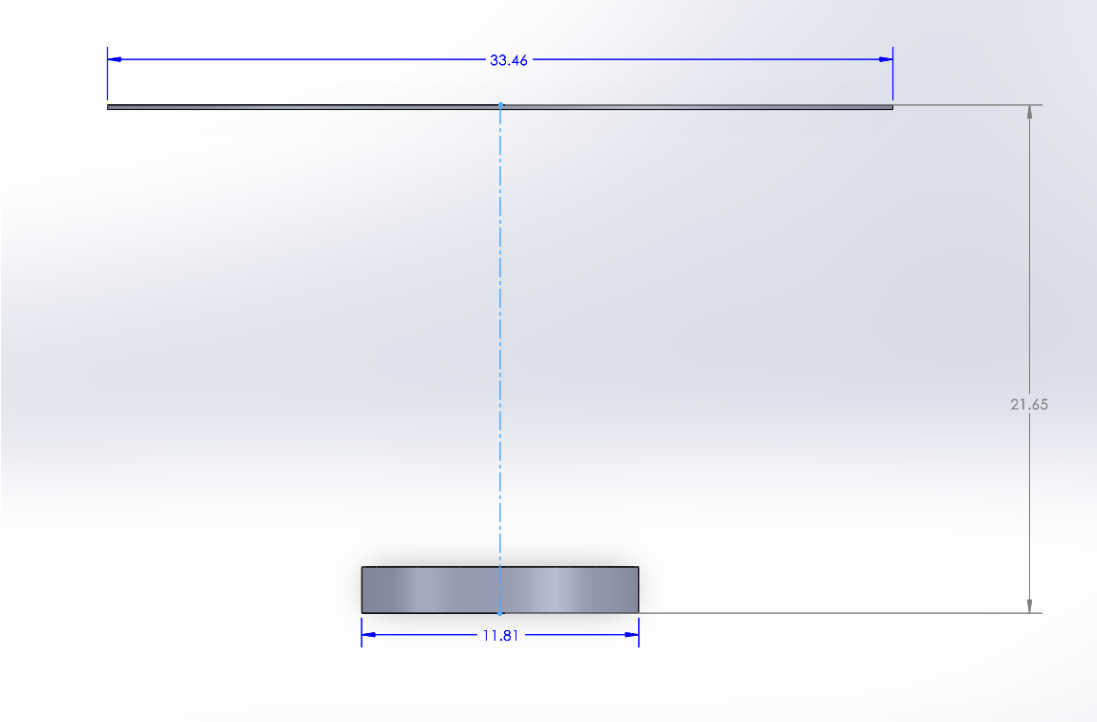


Figure 1: Measurements in inches, pounds, second.

The measurements converted to units of g, mm, second would be ~**300 mm** on the top ring and ~**850 mm** for the base ring, and height length of about **550 mm**.

The CAD development for the Tiny House support stand was guided by specific design and manufacturing requirements, as shown in Figure 1. The process began by importing the fixed mounting ring and ground base pad into the CAD environment. These parts were treated as fixed elements that could not be changed, ensuring that all design concepts followed the required mounting and ground contact rules.

The CAD models were created using parametric frameworks, which allowed quick adjustments to important dimensions such as strut angles, wall thickness, and base size. Each design was developed to reduce weight while maintaining strength and stiffness. Features like tapered profiles helped shorten load paths and improve structural performance, while triangulated members and ring stiffeners were added to resist lateral forces.

Dimensioned sketches and mass property checks were carried out in CAD to confirm that the design stayed below the 120 N weight limit. The final concept was then refined with detailed meshing and prepared for finite element analysis (FEA). This ensured that the CAD model was ready for simulation and suitable for manufacturing in aluminum using powder bed fusion.

Therefore, the criteria for selecting the part for simulation is about choosing the part with the least amount of weight. The benefits could be due to cost-efficiency when taking it for mass additive manufacturing. Therefore, the conceptual design as depicted in **Figure 2** is the least weight calculated amongst the other designs.

Axial Deadload of 2000N Validation

$$FoS = \frac{\sigma_{yield}}{\sigma_{max,\,von\,Mises}}$$

Simulation Result:

$$\sigma_{max,\,von\,Mises} \approx 148.5\,MPa$$

FoS for AlSi10Mg ($\sigma_{yield} \approx 230 - 260\,MPa$):

$$FoS \approx \frac{230 - 260}{14.85} \approx 1.55 - 1.75$$

Lateral Load of 500N Validation

$$FoS = \frac{\sigma_{yield}}{\sigma_{max,\,von\,Mises}}$$

Simulation Result:

$$\sigma_{max,\,von\,Mises} \approx 700\ MPa$$

FoS for AlSi10Mg ($\sigma_{yield} \approx 230 - 260\ MPa$):

$$FoS \approx \frac{230 - 260}{233\ -\ 408} \approx 0.56 - 1.11$$

Twisting Load of 100N-mm Validation

- Thin-walled polar moment (local top section)
  - $J = 2\pi r_t^3 t,\ r_t = \frac{3m}{2} = 0.150m, t = 1.0 * 10^{-4}m$
  - $J_{top} = 2.12 * 10^{-6}m^4$

- Polar moment over taper
  - $r^3 = \frac{r_t^4 - r_b^4}{4(r_t - r_b)}, r_b = \frac{5.50m}{2} = 0.425m$
  - $J_{avg} = 2\pi r^3 = 1.84 * 10^{-5}m^4$
- Applied torque
  - $T = 100N * mm = 0.1N * m$
- Shear stress at the top ring
  - $\tau_{op} = \frac{Tr_t}{J_{top}} = \frac{0.1*0.150}{2.12*10^{-6}} = 0.0071\ MPa$
- Global angle of twist
  - $\theta = \frac{TL}{GJ_{avg}} = \frac{0.1*0.55}{26*10^9*1.84*10^{-5}} \sim 1.15 * 10^{-7}rad$
- Tip deflection at the top radius:
  - $u = r_t\theta = 0.150 * 1.15 * 10^{-7} = 1.7 * 10^{-5}mm$

Frequency Resonance Validation

Using the final concept for this study, we model the stand as a tapered cantilever with linear diameter change from the base to the top.

- Top diameter, $D_t = 300mm$
- Base diameter, $D_b = 850mm$
- Height length, $L = 550mm$
- Wall thickness, $t = 0.1\ mm$
- Mass of part (aluminum density), $m_{eff} = 5.433kg$
- Young's Modulus, $E = 70\ GPa$
- Inertia, $I_{avg} = \frac{\pi}{32} * t * \frac{D_t^4 - D_b^4}{D_t - D_b} \sim 9.17 * 10^{-6} m^4$
- Equivalent tip stiffness:
  - $k_{eq} = \frac{2EI_{avg}}{L^3} = 1.14 * 10^7 \frac{N}{m}$
- Effective modal mass (first mode):
  - Mode 1, K1 = 1.87

$$Frequency(f_n) = \frac{1}{2\pi}\sqrt{\frac{k_{eq}}{m_{eff}}}$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{1.14 * 10^7}{5.433}} \sim 230\ Hz$$

Verifying with FEA:

FEA (Abaqus): ~ 200.3 Hz

$$\%error = \frac{Theoretical - FEA}{Theoretical} = \frac{|230 - 200.3|}{230} = 12.91\%$$

FEA (Fusion360) ~ 211.94 Hz

$$\%error = \frac{Theoretical - FEA}{Theoretical} = \frac{|230 - 211.94|}{230} = 7.85\%$$

Conclusions/Recommendations

The axial dead load FoS hand calculations of **1.55-1.75** under axial **2000 N**, the design meets static yield strength of the material as FoS greater than 1 means that the structure or component can withstand more load than it's expected to experience in service.

The lateral load FoS calculations of **0.56-1.11** under lateral load of **500 N**, the design does not meet static yield strength of the material as FoS less than 1 means that the structure or component will withstand the load than it's expected to experience in service. Local stress hotspots near the ring indicate opportunities for fillet growth, thickness of tuning, or load-spread features to further raise FoS without significantly increasing mass of the design.

The torsional hand calculations corroborate the FEA, indicating very low stress and deflection under **100 N-mm**. The design meets torsional stiffness and strength requirements. The hand calculation predicts the top ring's shear stress of about **0.0071 MPa**, which is within the same order of magnitude as the FEA Abaqus value of **0.00445 MPa**. The tip deflection of $\mathbf{1.7 * 10^{-5}}\boldsymbol{mm}$ is close to the FEA Abaqus result of $\mathbf{3.7 * 10^{-5}}\boldsymbol{mm}$. The small differences are expected due to tapering, discrete features near the mounting ring, and local stress concentrations.

When verifying the resonance hand calculations against the values generated from finite element analysis software such as Abaqus and Fusion360, the measured values and the simulated values do not deviate significantly since the values deviate as much as **~13%** from the theoretical. The purpose of conducting resonance (modal) analysis for the structure is to evaluate the stiffness of the structure when subjected to vibrations being applied from external loads. Since the design requirement specifies that the lowest natural frequency must be around 200 Hz, our simulated values of about ~200 - 212 Hz confirms that the structure satisfies the resonance requirements and is considered a sound design.

Future Work

While the final product meets all needed requirements, there are several areas that could be changed in future iterations to increase performance, reduce weight, and improve optimization. From the base design, some of the supports and struts could be thinned to further decrease the weight. Low stress areas can be examined to see if material can be removed or modified while maintaining the required strength. The addition of fillets at the intersections could help reduce load concentrations. Thickening areas of increased stress and reducing it in others may also lead to ways to increase the stands' durability while reducing the weight. The building material could also be considered. If weight is more important than cost, there are some materials that could be used. Titanium Ti-6Al-4V offers nearly triple the yield strength at similar mass which would allow for a reduction in the required number of supports or thickness. Since this is a product using additive manufacturing, the differences in the heat treatment cycle should also be considered; some can lead to increased yield strength or other benefits.

Besides the physical design of the stand, the interaction between the stand and the tiny house can be taken into account for future iterations. In the real world how well will the stand interact with its environment? What can come from earthquakes, soil shifting underneath, or vibrations coming from inside the tiny house. Studies can be done to see how these situations affect the stability of the stand on different types of ground, for example on grass, gravel, soil, or concrete. Would adjustability be useful to add to the stand in case of uneven ground? What is the expected lifespan of these stands, where fatigue is first noticed, and how corrosion and wear show up and affect the stand can all be investigated and improved upon. Another area that can be checked is how the layout of the interior of the tiny house affects the stands. Is the corner with the kitchenet causing more wear and fatigue on the stand nearest it or is it the sleeping area. How the designed tiny house stand interacts with the target product could change how its design is changed in the future.

These topics could all help refine the support stand beyond its current requirements. Further reducing its weight, achieving a higher performance, and improving long-term durability. Studying its use and how it interacts with a tony house can lead to future iterations of the stand to be stronger and more reliable in real world conditions. Continued optimization of the product through geometric refinement, real world validation and material selection can guide the design to an optimal field-ready support stand.

References

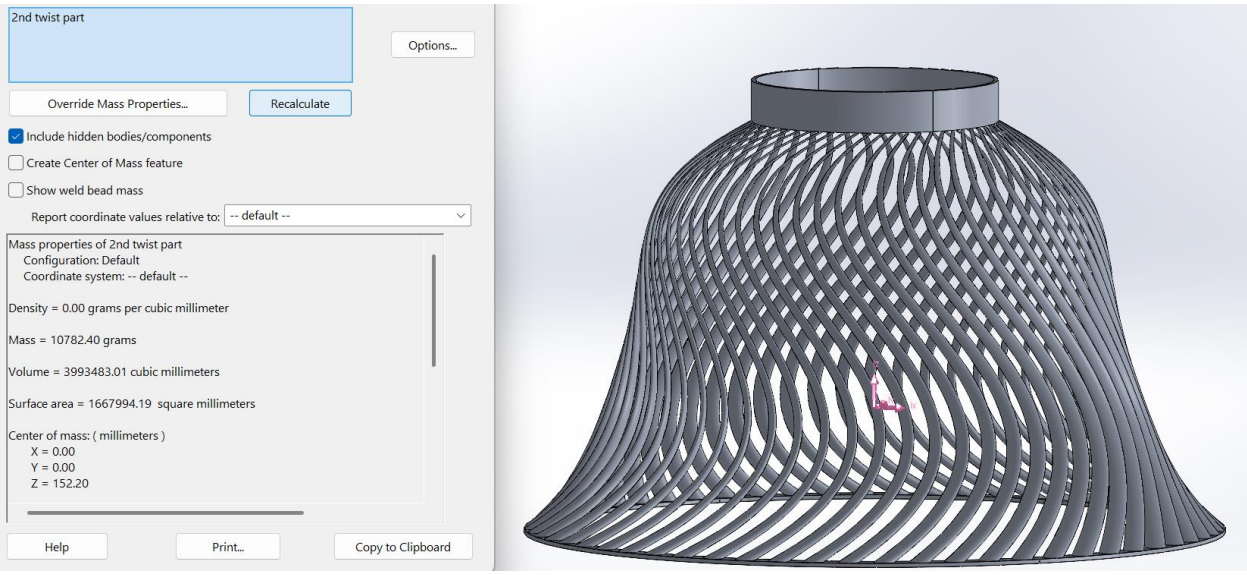
[1] V. Kvvssn et al., “Recent progress in additive manufacturing of 7xxx aluminum alloys - the International Journal of Advanced Manufacturing Technology,” SpringerLink, https://link.springer.com/article/10.1007/s00170-025-15415-z

[2] Chalmers, https://research.chalmers.se/publication/541619/file/541619_Fulltext.pdf

[3] Matzesl, “Additive manufacturing part cost and pricing,” AMPOWER, https://ampower.eu/insights/additive-manufacturing-part-cost-and-pricing/

Appendix

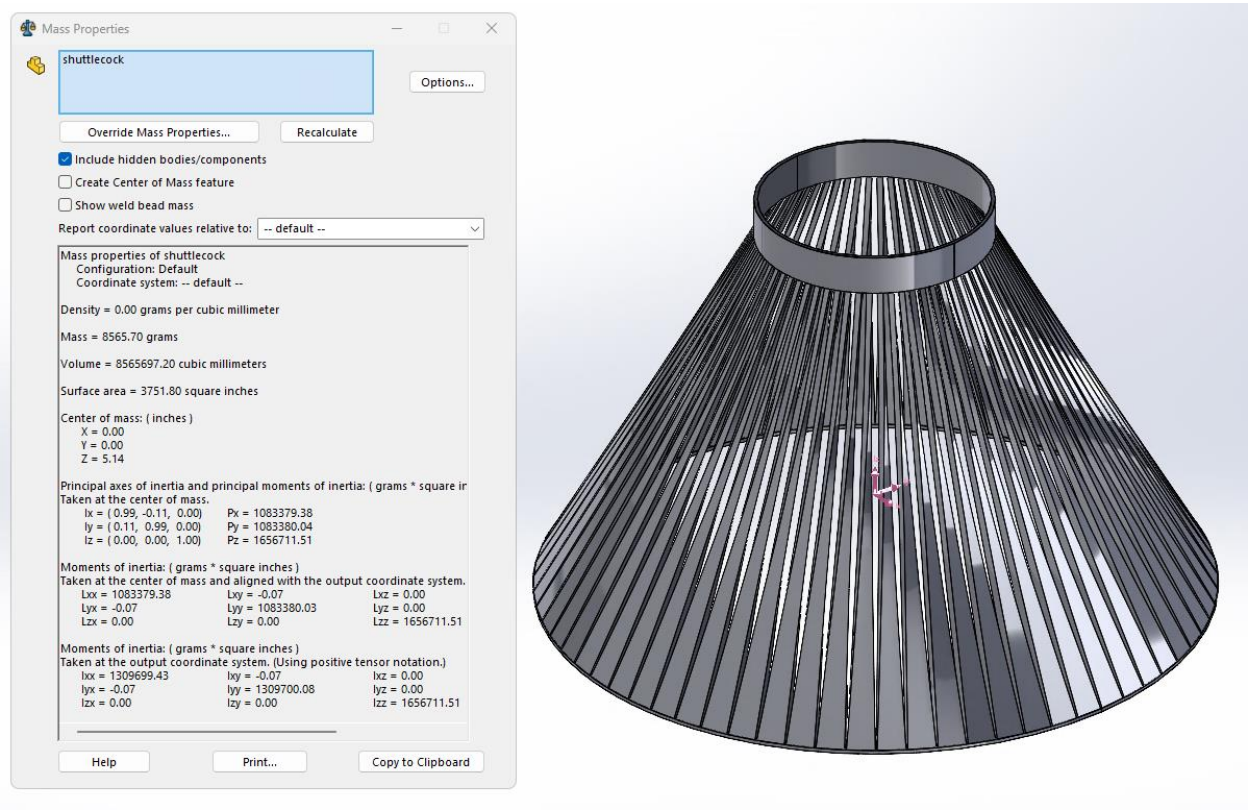
Concept 1: Uniform Conical shell



The measured weight is around $10782.40\ g \rightarrow 10.782\ kg$

$$10.782\ kg * 9.81 \frac{m}{s^2} = 105.77\ N < 120\ N\ weight\ requirement.$$

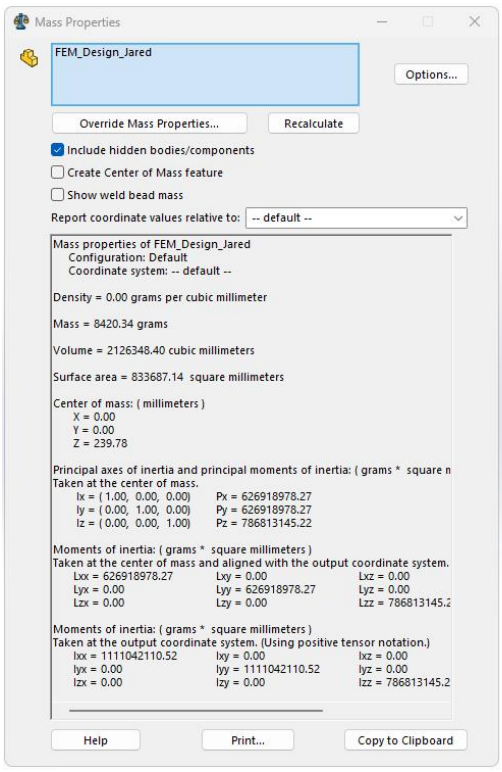
Concept 2: Conical shell with multiple spaced ribs

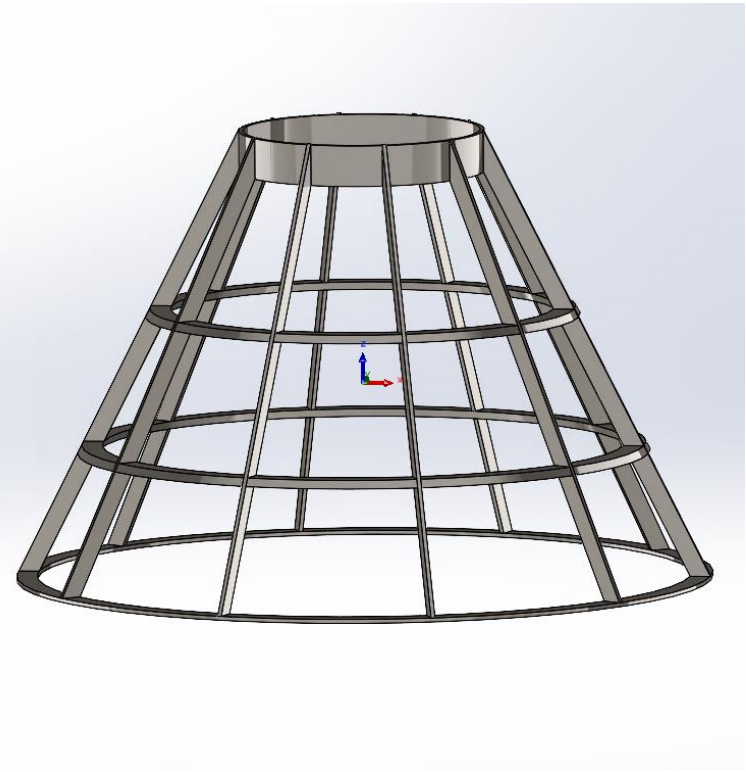


The measured weight is around $8565.70\ g \rightarrow 8.565\ kg$

$$8.565\ kg * 9.81 \frac{m}{s^2} = 84.022\ N < 120\ N\ weight\ requirement.$$

Concept 3: Conical Frames with Ribs

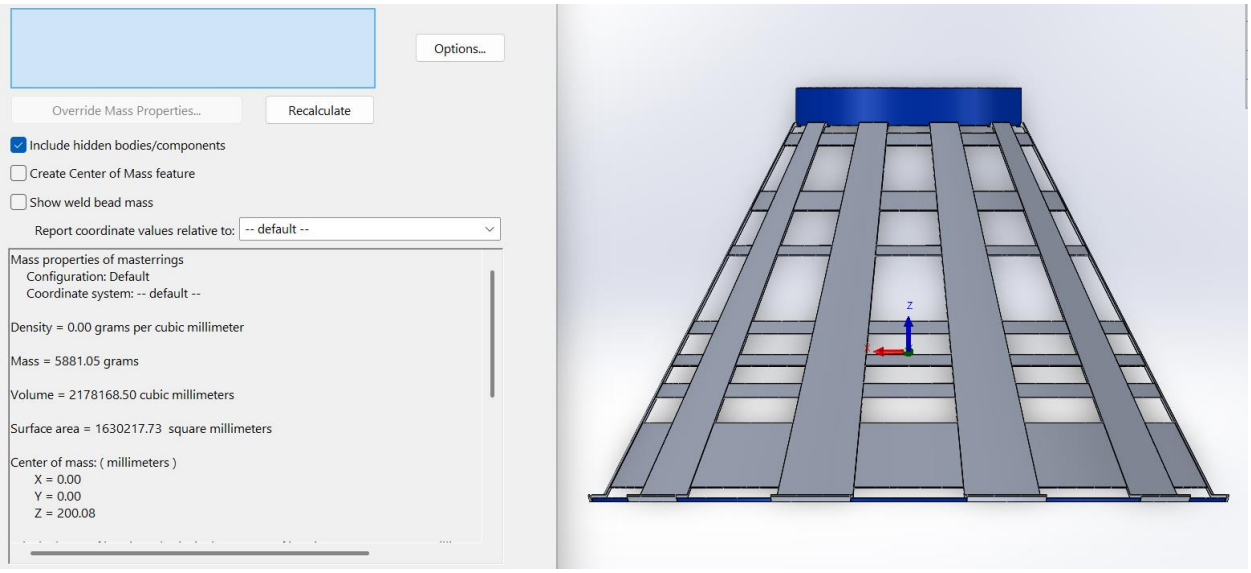




The measured weight is around $8420.34\ g \rightarrow 8.420\ kg$

$$8.420\ kg * 9.81 \frac{m}{s^2} = 82.60\ N < 120\ N\ weight\ requirement.$$

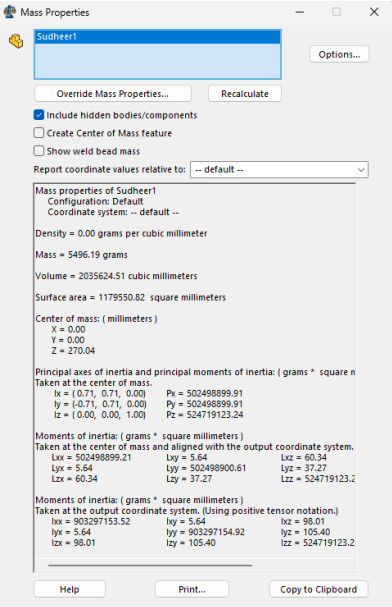
Concept 4: Ribbed conical shell structure

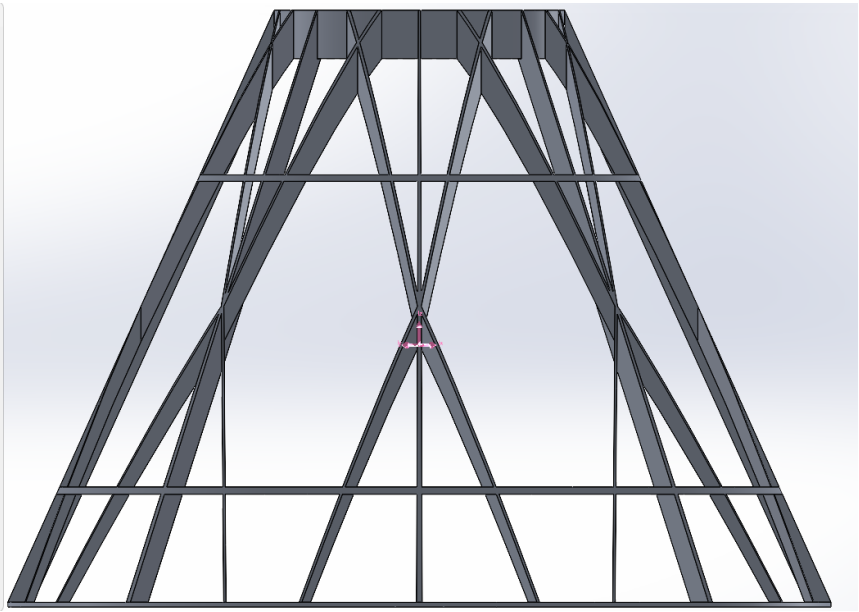


The measured weight is around 5881.05g→ $5.881g$

$$5.881\ kg * 9.81\frac{m}{s^2} = 57.69\ N < 120\ N\ weight\ requirement.$$

Concept 5: Tripod Truss design

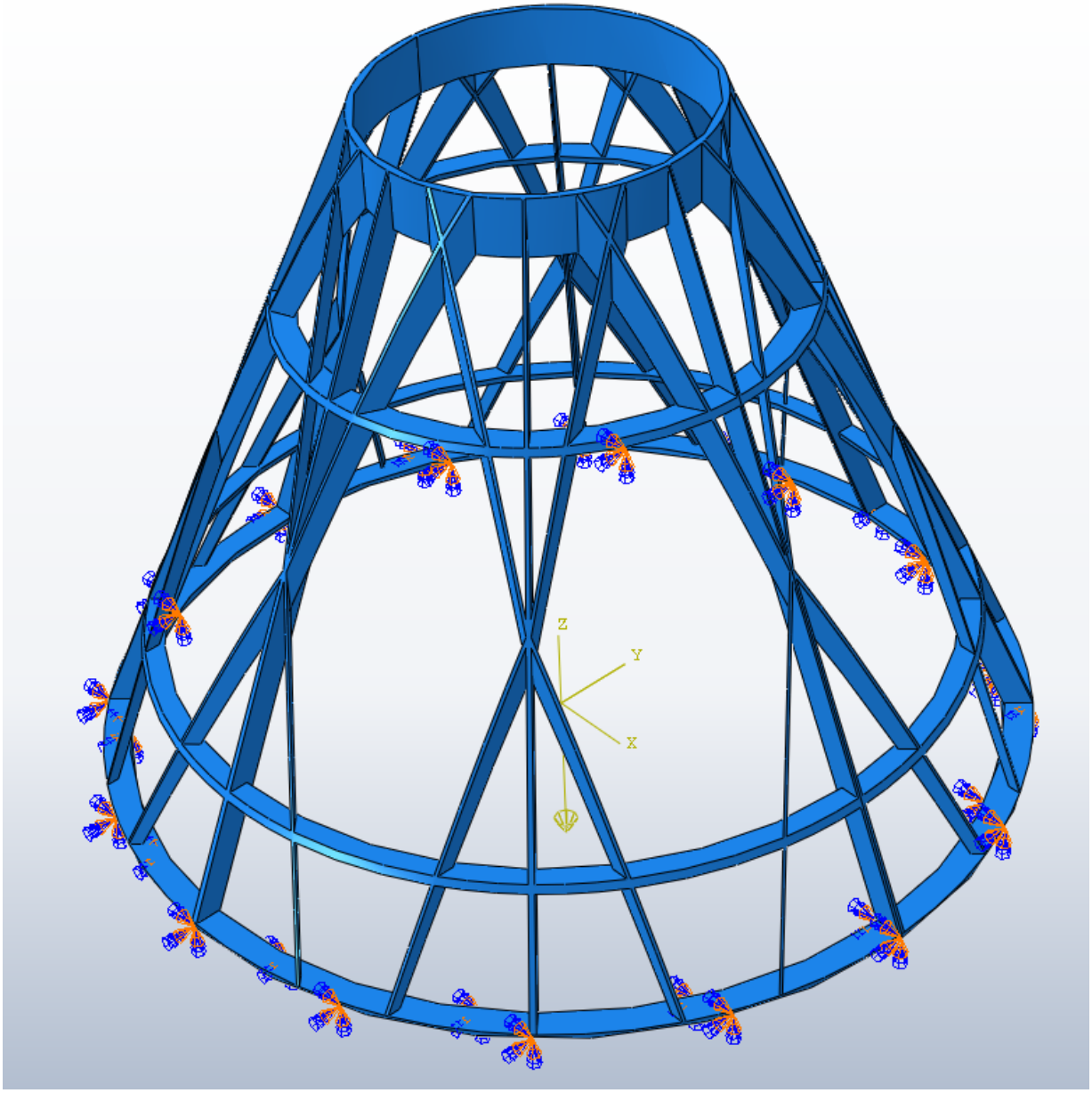


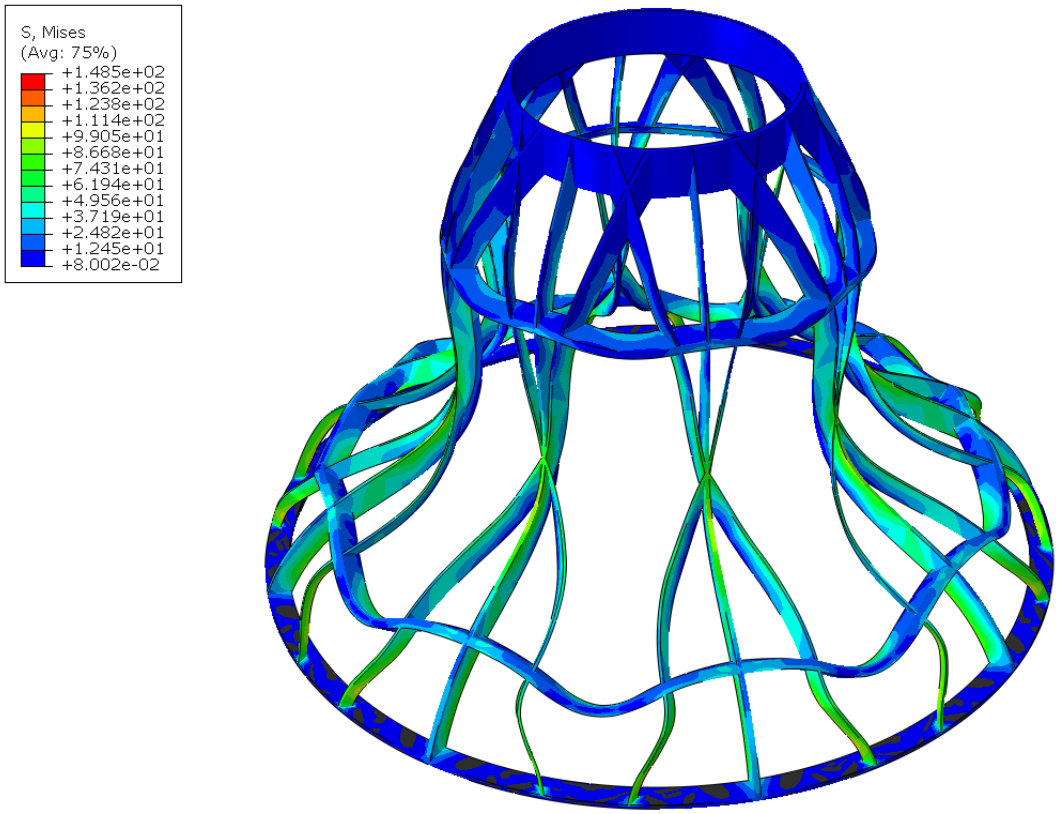


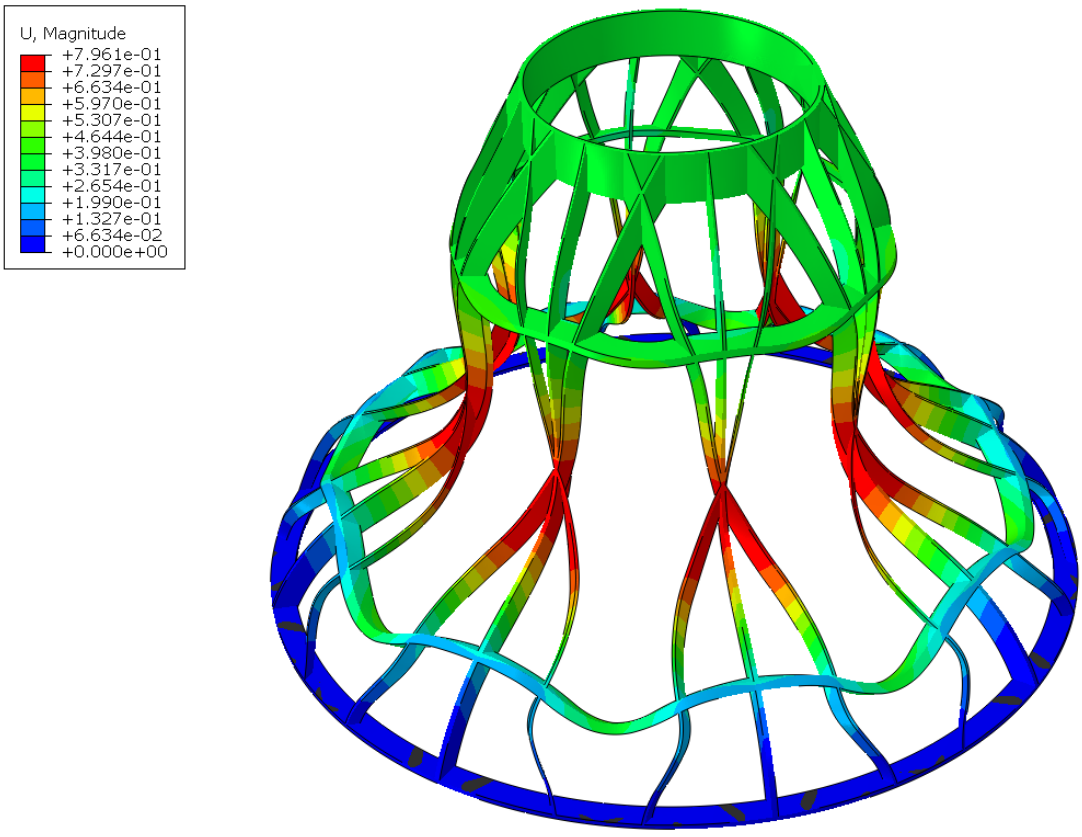
The measured weight is around $5.496\ kg\ (with\ Aluminum\ density)$

$$5.496\ kg * 9.81\frac{m}{s^2} = 53.91\ N < 120\ N\ weight\ requirement.$$

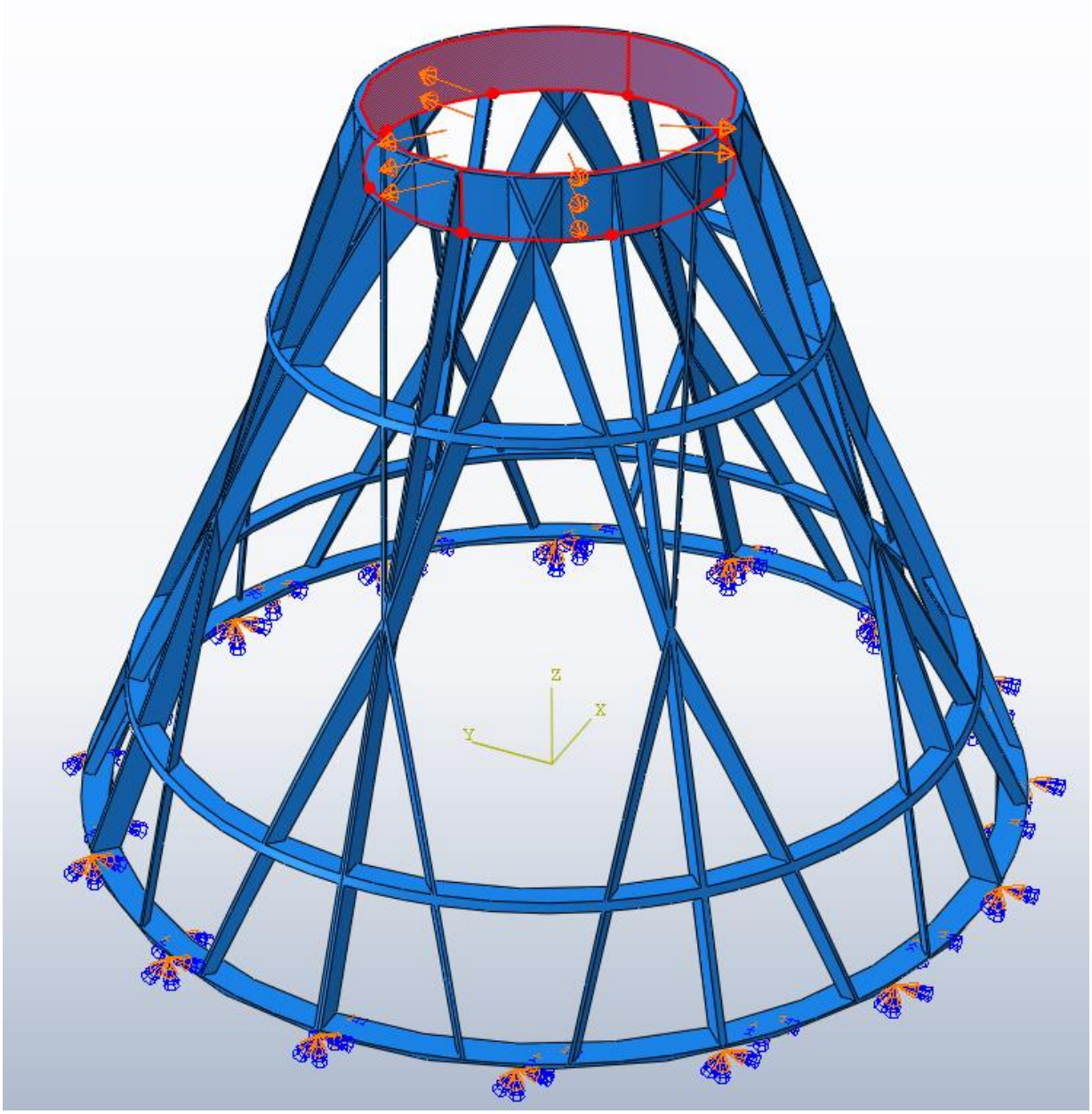
Axial Deadload of 2000 N Simulations

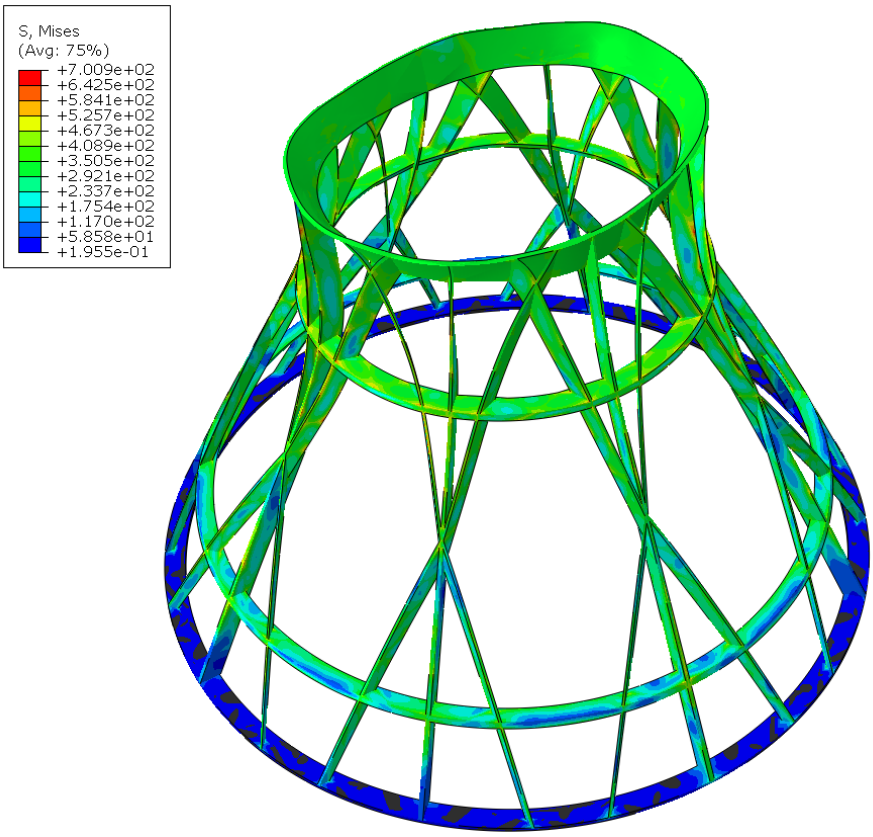


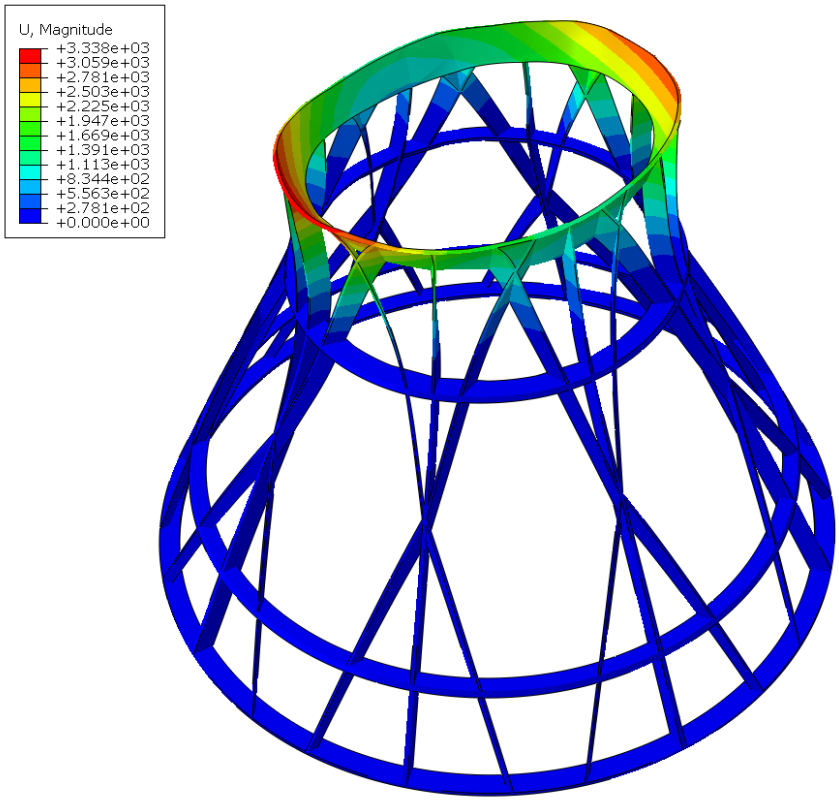

S, Mises
(Avg: 75%)
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+1.362e+02
+1.238e+02
+1.114e+02
+9.905e+01
+8.668e+01
+7.431e+01
+6.194e+01
+4.956e+01
+3.719e+01
+2.482e+01
+1.245e+01
+8.002e-02


U, Magnitude
+7.961e-01
+7.297e-01
+6.634e-01
+5.970e-01
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+4.644e-01
+3.980e-01
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+0.000e+00

Lateral Load of 500N Simulations

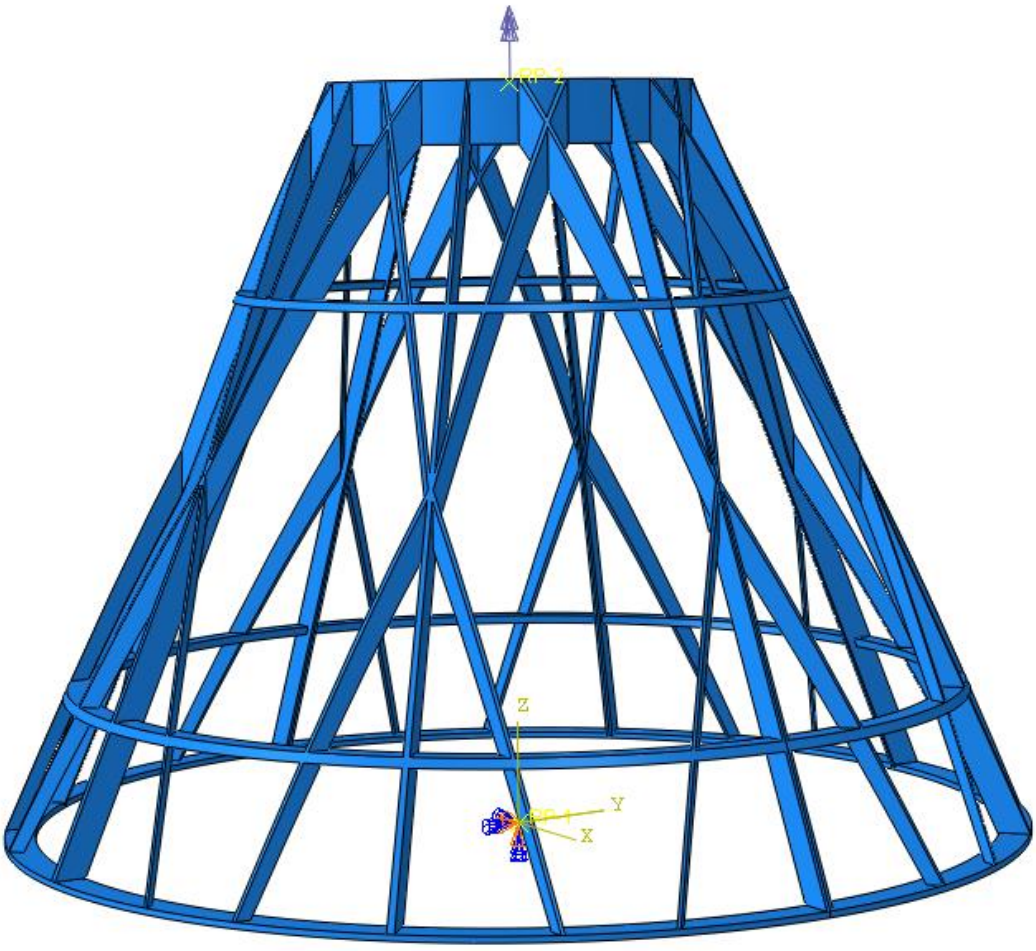


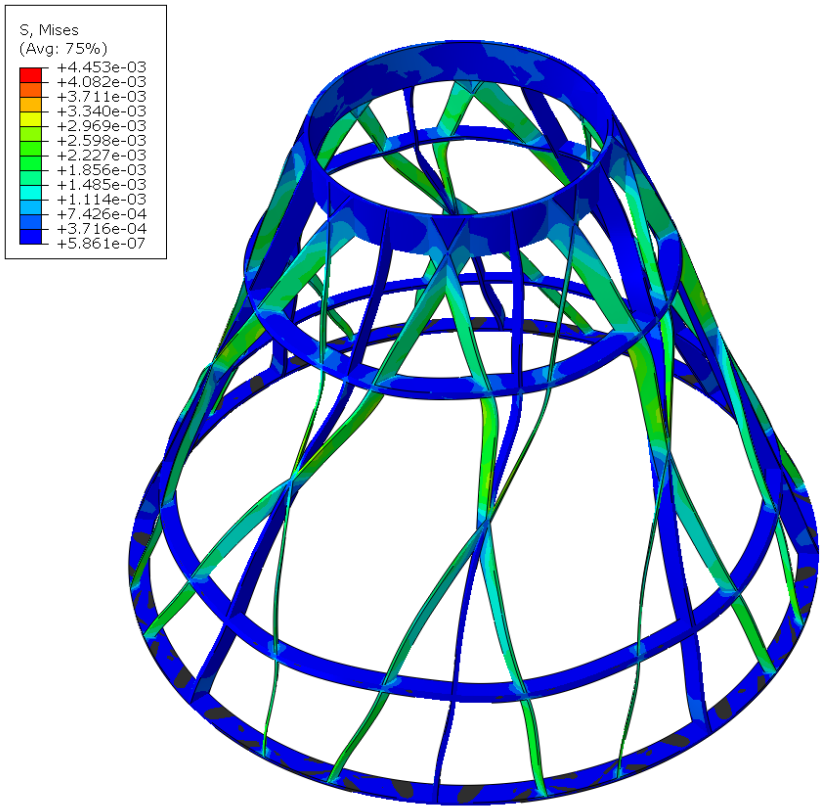

S, Mises
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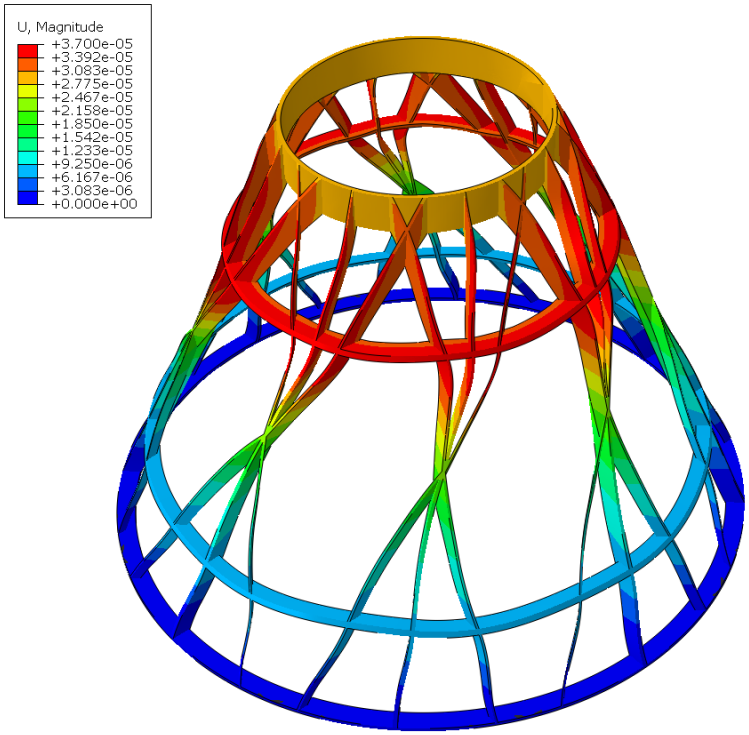

U, Magnitude
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+2.781e+03
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+1.947e+03
+1.669e+03
+1.391e+03
+1.113e+03
+8.344e+02
+5.563e+02
+2.781e+02
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Twisting Load of 100 N-mm Simulations

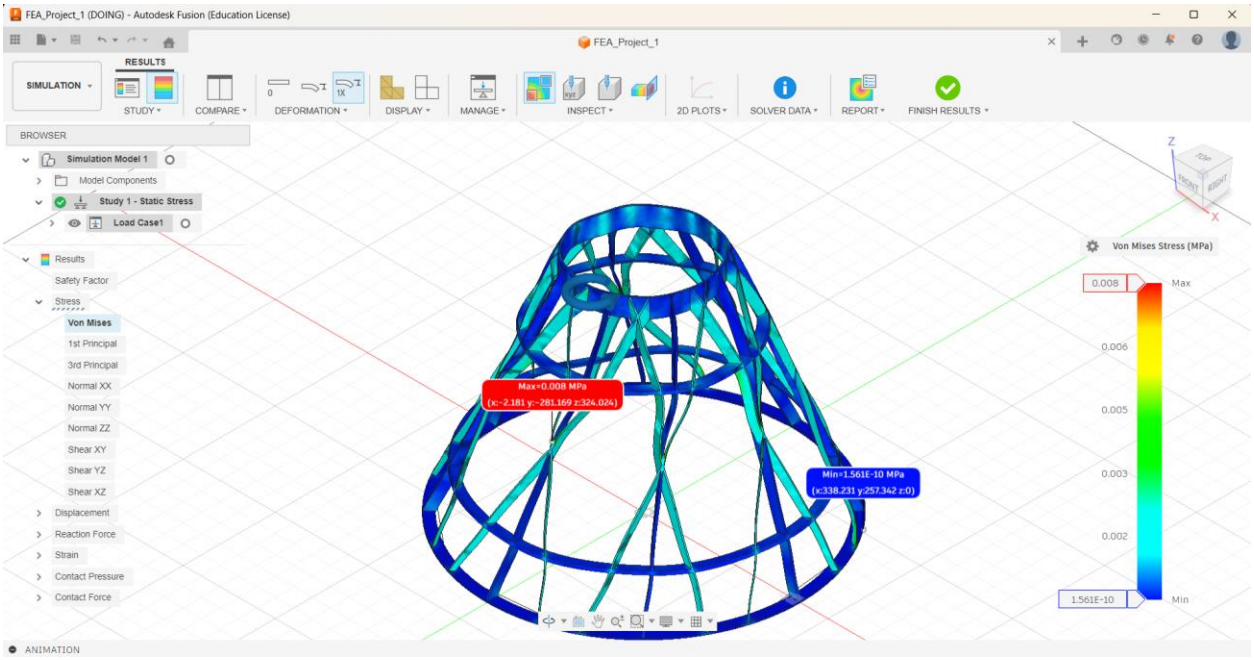
Abaqus:

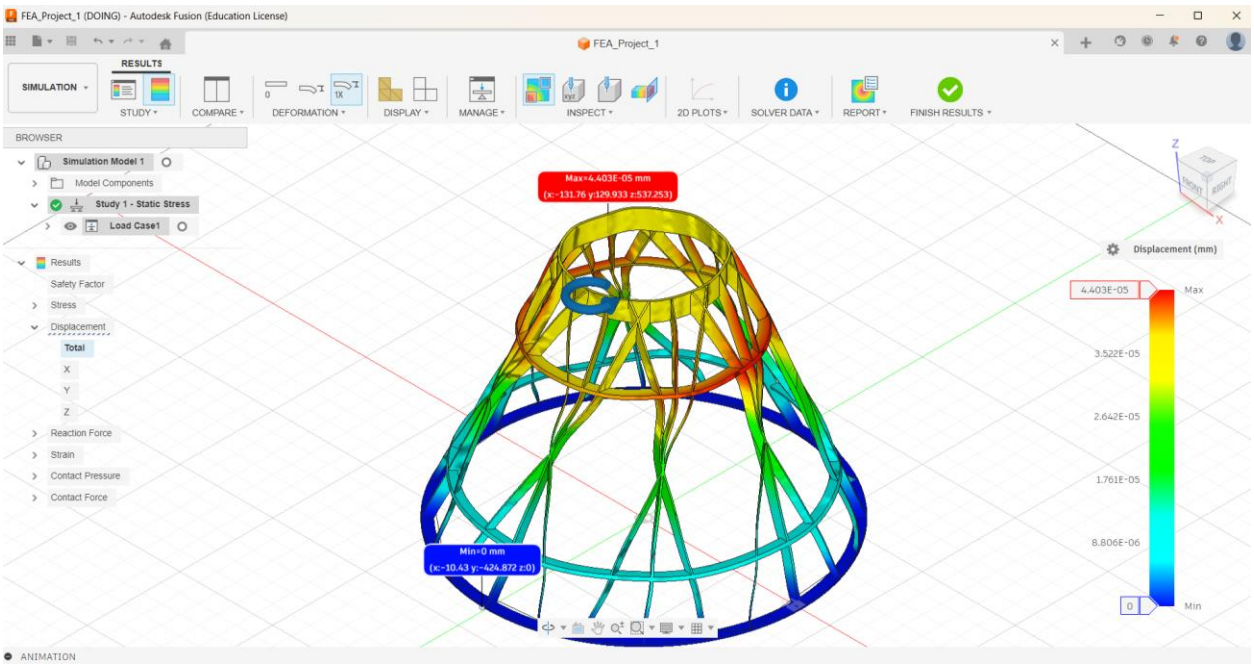



S, Mises
(Avg: 75%)
+4.453e-03
+4.082e-03
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+2.598e-03
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+1.114e-03
+7.426e-04
+3.716e-04
+5.861e-07


U, Magnitude
+3.700e-05
+3.392e-05
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+2.775e-05
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+1.850e-05
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Fusion360:


FEA_Project_1 (DOING) - Autodesk Fusion (Education License)
FEA_Project_1
SIMULATION
RESULTS
STUDY
COMPARE
DEFORMATION
DISPLAY
MANAGE
INSPECT
2D PLOTS
SOLVER DATA
REPORT
FINISH RESULTS
BROWSER
Simulation Model 1
Model Components
Study 1 - Static Stress
Load Case1
Results
Safety Factor
Stress
Von Mises
1st Principal
3rd Principal
Normal XX
Normal YY
Normal ZZ
Shear XY
Shear YZ
Shear XZ
Displacement
Reaction Force
Strain
Contact Pressure
Contact Force
Max=0.008 MPa
(x:-2.181 y:-281.169 z:324.024)
Min=1.561E-10 MPa
(x:338.231 y:257.342 z:0)
Von Mises Stress (MPa)
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Max
0.006
0.005
0.003
0.002
1.561E-10
Min
ANIMATION

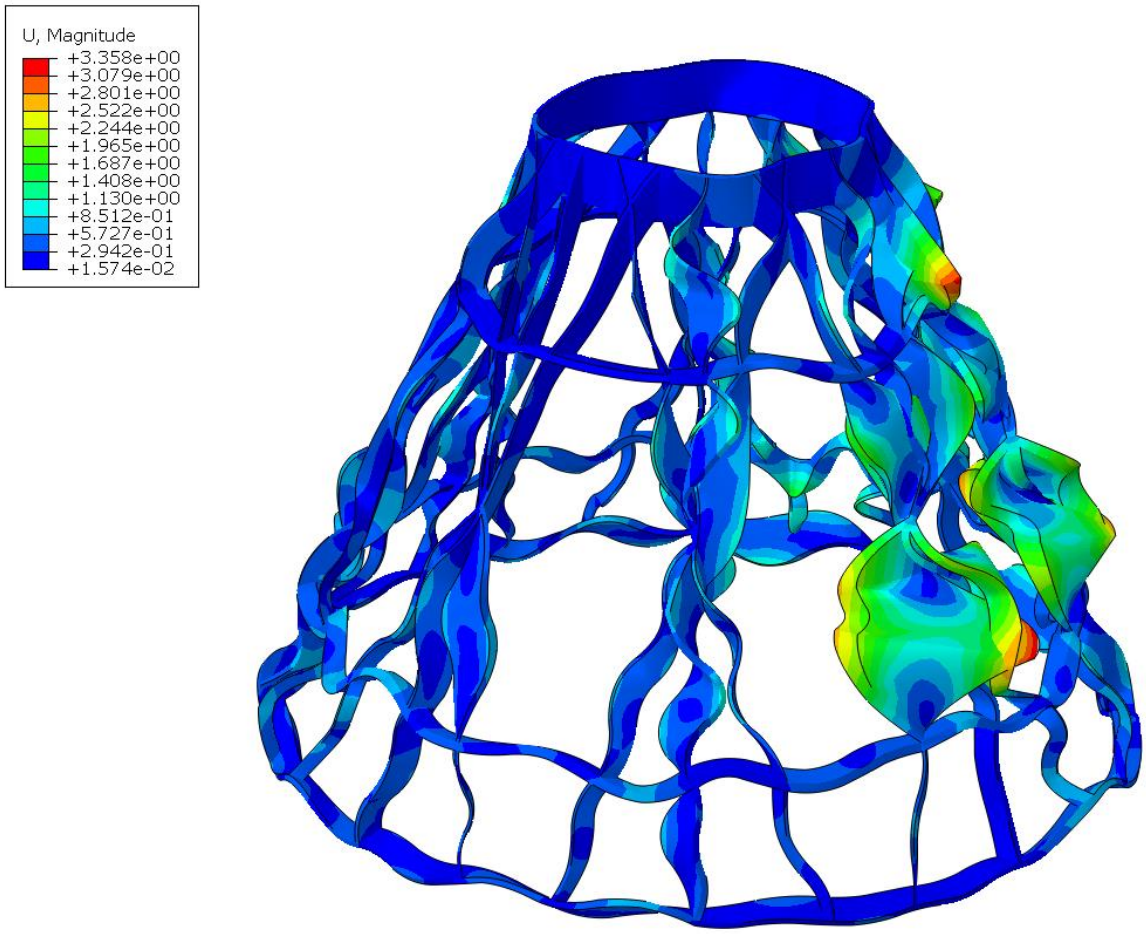

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FEA_Project_1
SIMULATION
RESULTS
STUDY
COMPARE
DEFORMATION
DISPLAY
MANAGE
INSPECT
2D PLOTS
SOLVER DATA
REPORT
FINISH RESULTS
BROWSER
Simulation Model 1
Model Components
Study 1 - Static Stress
Load Case1
Results
Safety Factor
Stress
Displacement
Total
X
Y
Z
Reaction Force
Strain
Contact Pressure
Contact Force
Max=4.403E-05 mm
(x:-131.76 y:129.933 z:537.253)
Min=0 mm
(x:-10.43 y:-424.872 z:0)
Displacement (mm)
4.403E-05
Max
3.522E-05
2.642E-05
1.761E-05
8.806E-06
0
Min
ANIMATION

Resonance Load Simulations

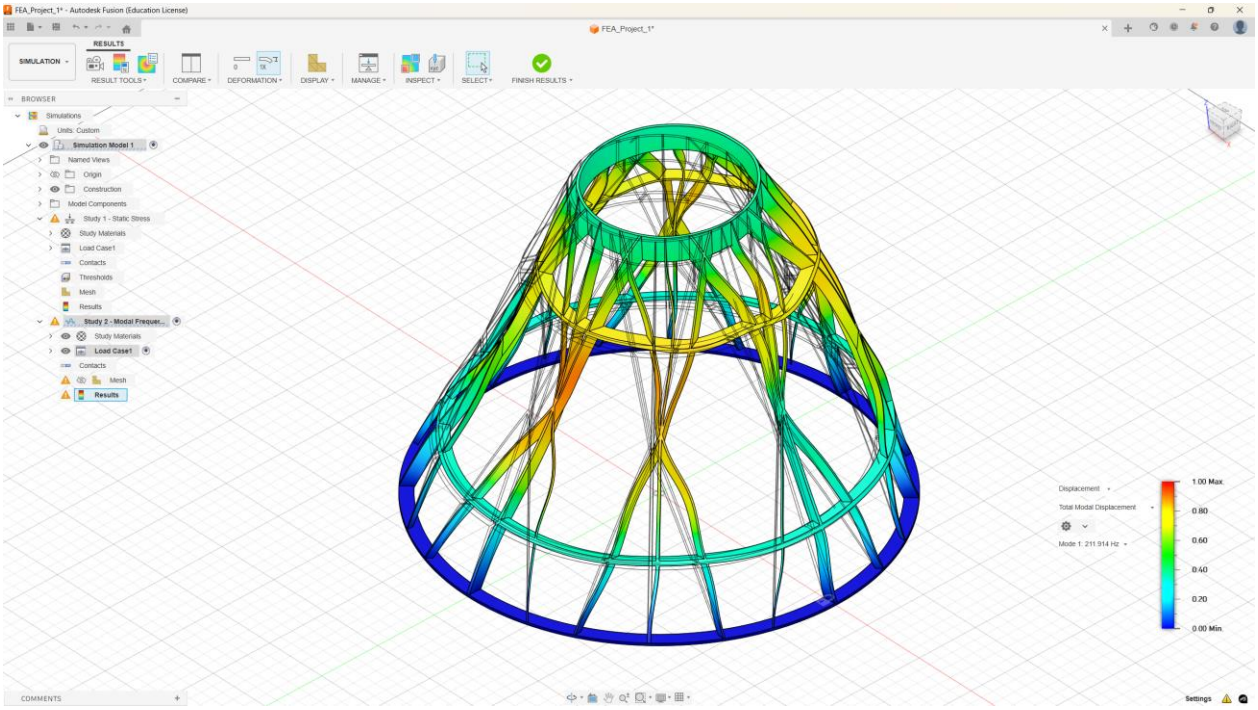
Abaqus:

Frame

Index	Description
0	Increment 0: Base State
1	Mode 1: Value = 1.58423E+06 Freq = 200.32 (cycles/time)



Fusion360:


FEA_Project_1* - Autodesk Fusion (Education License)
FEA_Project_1*
SIMULATION
RESULTS
RESULT TOOLS
COMPARE
DEFORMATION
DISPLAY
MANAGE
INSPECT
SELECT
FINISH RESULTS
BROWSER
Simulations
Units: Custom
Simulation Model 1
Named Views
Origin
Construction
Model Components
Study 1 - Static Stress
Study Materials
Load Case1
Contacts
Thresholds
Mesh
Results
Study 2 - Modal Freque...
Study Materials
Load Case1
Contacts
Mesh
Results
COMMENTS
Displacement
Total Modal Displacement
Mode 1: 211.914 Hz
1.00 Max
0.80
0.60
0.40
0.20
0.00 Min
Settings


Displacement
Total Modal Displacement
Mode 1: 211.914 Hz

Team Responsibilities

Hamid, Farris	• Formatted the Report Writing • Help Designed Concept 1 and Concept 2 models. • Did the Axial Deadload and Torsional Load Simulations in Abaqus for the Final Concept Design. • Wrote the PowerPoint Presentation Slides for the Simulations which I performed.
Smentkowski, Jared	– Formatted the PowerPoint – Helped design models
Kollu, Sudheer Kumar	• Did simulations on twisting and resonance • Coordinated on designing models
Arnab, Shams Fardous	• Helped Design models • Did twisting and resonance test on Abaqus. • Crosschecked simulations in Abaqus